M.A.M SCHOOL OF ENGINEERING

(An Autonomous Institution)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University)

Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

B.E. BIOMEDICAL ENGINEERING

CURRICULUM

(CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION)

I TO VIII SEMESTERS CURRICULUM

(Applicable for the students admitted from 2024-2025 onwards)

REGULATIONS 2024

M.A.M SCHOOL OF ENGINEERING

(AN AUTONOMOUS INSTITUTION)

REGULATIONS 2024

CHOICE BASED CREDIT SYSTEM

B. E. BIOMEDICAL ENGINEERING

I. PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

1. To enable the graduates to demonstrate their skills in design and develop medical devices for health care system through the core foundation and knowledge acquired in engineering and biology.
2. To enable the graduates to exhibit leadership in health care team to solve health care problems and make decisions with societal and ethical responsibilities.
3. To Carryout multidisciplinary research, addressing human healthcare problems and sustain technical competence with ethics, safety and standards.
4. To ensure that graduates will recognize the need for sustaining and expanding their technical competence and engage in learning opportunities throughout their careers.

II.PROGRAM OUTCOMES (POs)

PO1	**Engineering Knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem Analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6	**The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.**
PO7	**Environment and Sustainability: Understand the impact of the professional engineering solutionsin societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.**
PO8	**Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.**
PO9	**Individual and Team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.**
PO10	**Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.**
PO11	**Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.**
PO12	**Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.**

III.PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1	**To design and develop diagnostic and therapeutic devices that reduces physician burnout and enhance the quality of life for the end user by applying fundamentals of Biomedical Engineering.**
PSO2	**To apply software skills in developing algorithms for solving healthcare related problems in various fields of Medical sector.**
PSO3	**To adapt to emerging information and communication technologies (ICT) to innovate ideas and solutions for current societal and scientific issues thereby developing indigenous medical instruments that are on par with the existing technology**

CURRICULUM

M.A.M SCHOOL OF ENGINEERING

DEPARTMENT OF BIOMEDICAL ENGINEERING

REGULATIONS 2024

CHOICE BASED CREDIT SYSTEM

(Students admitted from the Academic Year 2024 – 25 onwards)

I TO VIII SEMESTERS CURRICULUM

Induction Program (Mandatory)	3 weeks duration
Induction program for students to be offered right at the start of the first year	• Physical activity • Creative Arts • Universal Human Values • Literary • Proficiency Modules • Lectures by Eminent People • Visits to local Areas • Familiarization to Dept. / Branch & Innovations

B.E. BIOMEDICAL ENGINEERING

SEMESTER I

S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24HS101	Communicative English	2	0	0	2	40	60	100	HS
2.	24HS102	Heritage of Tamil	1	0	0	1	40	60	100	HS
3.	24BS101	Matrices & Calculus	3	1	0	4	40	60	100	BS
4.	24ES101	Problem Solving and Python Programming	3	0	0	3	40	60	100	ES
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24BS103	Engineering Physics	3	0	2	4	50	50	100	BS
LABORATORY COURSES										
7.	24HS103	Communicative English Laboratory	0	0	2	1	60	40	100	HS
8.	24ES103	Problem Solving and Python Programming Laboratory	0	0	3	2	60	40	100	ES
9.	24ES106	Engineering Practices Laboratory	0	0	2	1	60	40	100	ES
10.	24ES107	Workshop Practices Laboratory	0	0	2	1	60	40	100	ES
		TOTAL	13	1	12	19				

S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
SEMESTER II										
THEORY COURSES										
1.	24HS201	தமிழரும் தொழில் நுட்பமும் / Tamils and Technology	1	0	0	1	40	60	100	HS
2.	24HS202	Professional English	2	0	0	2	40	60	100	HS
3.	24BS201	Transforms and Partial Differential Equations	3	1	0	4	40	60	100	BS
4.	24ES201	Design Thinking	2	0	0	2	40	60	100	ES
THEORY COURSES WITH LABORATORY COMPONENT										
5.	24BS203	Chemistry for Engineers	3	0	2	4	50	50	100	BS
6.	24ES213	Electrical Engineering for medical applications	3	0	2	4	50	50	100	ES
7.	24ES203	Circuit Analysis for Biomedical Engineers	3	0	2	4	50	50	100	ES
LABORATORY COURSES										
8.	24ES205	Engineering Drawing	0	0	4	2	60	40	100	ES
9.	24TP201	Aptitude Skills and Communication skills I	0	0	2	1	100	-	100	EEC
		TOTAL	17	1	12	23				

SEMESTER III										
THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1.	24BS301	Statistics and Numerical Methods	3	1	0	4	40	60	100	BS
2.	24ECE01	Fundamentals of Electronic Devices and Circuits	3	0	0	3	40	60	100	PC
3.	24BM301	Biological sciences	3	0	0	3	40	60	100	PC
4.	24CS302	Object Oriented Programming	3	0	0	3	40	60	100	PC
THEORY COURSES WITH LABORATORY COMPONENT										
5.	24BM302	Human Anatomy and Physiology	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
6.	24CS304	Object Oriented Programming Lab	0	0	4	2	60	40	100	PC
7.	24BM303	Biological Sciences Laboratory	0	0	4	2	60	40	100	PC
8.	24ECE02	Electronic Devices and Circuits Lab	0	0	4	2	60	40	100	PC
9.	24TP301	Aptitude Skills and Communication skills I	0	0	2	1	100	-	100	EEC
		TOTAL	15	1	16	24				

SEMESTER IV										
THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1.	24BS401	Random process and linear algebra	3	1	0	4	40	60	100	BS
2.	24ECE03	Analog and Digital Integrated Circuits	3	0	0	3	40	60	100	PC
3.	24BM401	Biomedical Instrumentation	3	0	0	3	40	60	100	PC
4.	24MC401	Environmental Sciences	2	0	0	0	40	60	100	MC
5	24MC402	Disaster Management and Preparedness	2	0	0	2	40	60	100	MC
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24BM402	Biosensors and Measurements	3	0	2	4	50	50	100	PC
7.	24ECE04	Bio-Signal Processing	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
8.	24ECE05	Analog and Digital Integrated Circuits Lab	0	0	4	2	60	40	100	PC
9.	24BM403	Biomedical Instrumentation Lab	0	0	4	2	60	40	100	PC
10.	24TP401	Aptitude Skills III & Technical Skills	0	0	2	1	100	-	100	EEC
		TOTAL	18	1	14	25				

SEMESTER V

S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24BM501	Diagnostic and Therapeutic Equipment	3	0	0	3	40	60	100	PC
2.	24BM502	Biocontrol System	3	0	0	3	40	60	100	PC
3.		Professional Elective-I	3	0	0	3	40	60	100	PE
4.		Professional Elective-II	3	0	0	3	40	60	100	PE
5.		Open Elective-I	3	0	0	3	40	60	100	OE
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24BM503	Embedded System and IOMT	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7.	24BM504	Diagnostic and Therapeutic Equipment Laboratory	0	0	4	2	60	40	100	PC
8.	24TP501	Aptitude Skills IV & Technical Skills II	0	0	2	1	100	-	100	EEC
		TOTAL	18	0	8	22				

SEMESTER VI										
THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1.	24HS601	Principles of Management	3	0	0	3	40	60	100	HS
2.		Professional Elective – III	3	0	0	3	40	60	100	PE
3.		Professional Elective – IV	3	0	0	3	40	60	100	PE
4.		Open Elective – II	3	0	0	3	40	60	100	OE
THEORY COURSES WITH LABORATORY COMPONENT										
5.	24BM601	Artificial Intelligence and Machine Learning in Health Care	3	0	2	4	50	50	100	PC
6.	24BM602	Medical Image Processing	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7.	24TP601	Aptitude Skills V & Technical Skills III	0	0	2	1	100	-	100	EEC
		TOTAL	18	0	10	21				

SEMESTER VII										
THEORY COURSES										
S.NO	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
1.	24HS701	Human Values and Ethics	3	0	0	3	40	60	100	HS
2.	24BM701	Healthcare Analytics	3	0	0	3	40	60	100	PC
3.		Professional Elective – V	3	0	0	3	40	60	100	PE
4.		Open Elective – III	3	0	0	3	40	60	100	OE
LABORATORY COURSES										
5.	24ES701	Hospital Training*	0	0	4	2	100	0	100	EEC
6	24BM702	Mini Project	0	0	4	2	60	40	100	EEC
		TOTAL	12	0	4	16				

*To be undertaken during summer holidays

SEMESTER VIII										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
LABORATORY COURSES										
1.	24BM801	Project Work	0	0	20	10	60	40	100	EEC
		TOTAL	0	0	20	10	60	40	100	

PROFESSIONAL ELECTIVE COURSES						
S.No	Course Code	Course	L	T	P	C
PROFESSIONAL ELECTIVE I (HEALTHCARE MANAGEMENT)						
1.	24BMX01	Clinical Engineering	3	0	0	3
2.	24BMX02	Hospital Planning and Management	3	0	0	3
3.	24BMX03	Medical waste management	3	0	0	3
4.	24BMX04	Economics and Management for Engineers	3	0	0	3
5.	24BMX05	Biostatistics	3	0	0	3
6.	24BMX06	Forensic Science in Healthcare	3	0	0	3
7.	24BMX07	Indian traditional medical systems	3	0	0	3
8.	24BMX08	Health care information systems	3	0	0	3
PROFESSIONAL ELECTIVE II (MEDICAL DEVICE INNOVATION AND DEVELOPMENT)						
9.	24BMX09	Foundation Skills in Integrated Product Development	3	0	0	3
10.	24BMX10	Medical Device Design	3	0	0	3
11.	24BMX11	Patient Safety, Standards and Ethics	3	0	0	3
12.	24BMX12	Medical Device Regulations	3	0	0	3
13.	24BMX13	Medical Innovation and Entrepreneurship	3	0	0	3
14.	24BMX14	Rapid Prototyping	3	0	0	3
15.	24BMX15	Radiological equipment	3	0	0	3
16.	24BMX16	Pattern Recognition and Neural Networks	3	0	0	3

S.No	Course Code	Course	L	T	P	C
PROFESSIONAL ELECTIVE III (ADVANCED HEALTHCARE DEVICES)						
17.	24BMX17	Bio MEMS	3	0	0	3
18.	24BMX18	Critical Care Equipment	3	0	0	3
19.	24BMX19	Human Assist Devices	3	0	0	3
20.	24BMX20	Healthcare in 4.O	3	0	0	3
21.	24BMX21	Robotics in Healthcare	3	0	0	3
22.	24BMX22	Analytical Instrumentation	3	0	0	3
23.	24BMX23	Biomaterials and Artificial organs	3	0	0	3
24.	24BMX24	Bio Micro Fluidic Devices	3	0	0	3
PROFESSIONAL ELECTIVE IV (COMMUNICATION)						
25.	24BMX25	Communication Systems	3	0	0	3
26.	24BMX26	Wearable Devices	3	0	0	3
27.	24BMX27	Body Area Networks	3	0	0	3
28.	24BMX28	Virtual Reality and Augmented Reality in Healthcare	3	0	0	3
29.	24BMX29	Telehealth Technology	3	0	0	3
30.	24BMX30	Medical Informatics	3	0	0	3
31.	24BMX31	Medical Robotics	3	0	0	3
32.	24BMX32	Rehabilitation Engineering	3	0	0	3
PROFESSIONAL ELECTIVE V (SIGNAL AND IMAGE PROCESSING,RECENT TECHNOLOGIES)						
33.	24BMX33	Computer Vision	3	0	0	3
34.	24BMX34	Speech and Audio Signal Processing	3	0	0	3
35.	24BMX35	Brain Computer Interface and Applications	3	0	0	3
36.	24BMX36	Medical Imaging Systems	3	0	0	3
37.	24BMX37	Biometric Systems	3	0	0	3
38.	24BMX38	Medical Nanotechnology	3	0	0	3
39.	24BMX39	Medical Optics	3	0	0	3
40.	24BMX40	Biomaterials and characterization	3	0	0	3

OPEN ELECTIVES										
S.No	Code No	Course	L	T	P	C	Maximum Marks			Cat.
							CA	ES	Total	
1	24BMY01	Lifestyle Diseases	3	0	0	3	40	60	100	OE
2	24BMY02	Neural Engineering	3	0	0	3	40	60	100	OE
3	24BMY03	Biotechnology for Waste Management	3	0	0	3	40	60	100	OE
4	24BMY04	Critical Care and Operation Theatre Equipment	3	0	0	3	40	60	100	OE
5	24BMY05	Analytical Instrumentation	3	0	0	3	40	60	100	OE
6	24BMY06	Medical Imaging systems	3	0	0	3	40	60	100	OE
7	24BMY07	Project Report Writing	3	0	0	3	40	60	100	OE
8	24BMY08	Entrepreneurship Development	3	0	0	3	40	60	100	OE
9	24BMY09	Drinking Water Supply and treatment	3	0	0	3	40	60	100	OE
10	24BMY10	Drone technologies	3	0	0	3	40	60	100	OE
11	24BMY11	Urban Agriculture	3	0	0	3	40	60	100	OE

S.No.	Category	Credits Per Semester								Total Credit	Credits in %
		I	II	III	IV	V	VI	VII	VIII		
1	HS	5	3				3	3		14	8.5%
2	BS	8	8	4	4					24	14.6%
3	ES	7	12							19	11.6%
4	PC			19	18	12	8	3		60	36.6%
5	PE					6	6	3		15	9.1%
6	OE					3	3	3		9	5.5%
7	MC				2					2	1.2%
8	EEC		1	1	1	1	1	4	10	19	11.6%
Total		20	24	24	25	22	21	16	10	162	100%

HS – Humanities and Social Science

BS –Basic Science

ES –Engineering Science

PC – Professional Core

PE –Professional Elective

OE–Open Elective

EEC – Employability Enhancement Course

MC – Mandatory course

CA – Continuous Assessment

ES – End Semester Examination